

Digital Signal Processing (EEE F434/ ECE F434)

Lab 11: Filter Design

Instructions:

1. Lab report to be submitted (on Nalanda) at the end of the lab hours (preferably), late submission will result in loss of marks, use the lab hours to finish the work
 2. All the tasks in each lab have to be in one main file, differentiate each question by commenting. Alternatively, you can zip all .m files together, name them appropriately and upload one zip/rar file. Make one Word/pdf file with all the results corresponding to the questions
 3. Plots should include x label, y label, legend, and titles
 4. File name: ID_Lab11.rar or zip **and** ID_Lab11.pdf
 5. If the report/code is found to be copied, zero marks will be awarded [**Ensure that you show the outputs of the task to the TAs and get them evaluated. No queries will be entertained post lab session.**]
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1. A speech signal with a bandwidth of 4 kHz is sampled at 8 kHz. The signal is corrupted by sinusoids with frequencies 1 kHz, 2 kHz, and 3 kHz.
 - a) Design an IIR filter using notch filter components that eliminate these sinusoidal signals.
 - b) Choose the filter's gain so that the maximum gain is equal to 1, and plot the log-magnitude response of your filter.
 - c) Load the **Handel** sound file in MATLAB [*Available in MATLAB is a short snippet of Handel's hallelujah chorus, which is a digital sound about 9 seconds long, sampled at 8192 sam/sec*], and add the preceding three sinusoidal signals to create a corrupted sound signal. Now, filter the corrupted sound signal using your filter and comment on its performance.
2. Design an analog Butterworth lowpass filter with a 0.25 dB or better ripple at 500 rad/sec and at least 50 dB of attenuation at 2000 rad/sec. Determine the system function in a rational function form. Plot the magnitude response, the log-magnitude response in dB, the phase response, the group delay, and the impulse response of the filter.
3. Design a bandpass filter using the Hamming window design technique. The specifications are
lower stopband edge: 0.3π
upper stopband edge: 0.6π , $A_s = 50$ dB
lower passband edge: 0.4π
upper passband edge: 0.5π , $R_p = 0.5$ dB
Plot the impulse response and the magnitude response (in dB) of the designed filter. Do not use the **fir1** function.
 R_p is the passband ripple in dB, and A_s is the stopband attenuation in dB

Hint: As we have not yet covered the windowing technique for digital filter design in lectures Hamming window is given by

$$w(n) = \begin{cases} 0.54 - 0.46 \cos\left(\frac{2\pi n}{M-1}\right), & 0 \leq n \leq M-1 \\ 0, & \text{otherwise} \end{cases}$$

- In MATLAB, $w = \text{hamming}(M)$ returns the M-point Hamming window function in array w.

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- Impulse response of ideal LPF is a sinc function, which needs to be truncated on both sides to obtain an FIR filter, this process of truncation is called windowing.
- In general, $h(n)$ can be thought of as being formed by the product of $h_d(n)$ and a window function $w(n)$ as: $h(n) = h_d(n)w(n)$, where $h_d(n)$ is ideal frequency response given as

$$\frac{\sin [\omega_c(n - \alpha)]}{\pi(n - \alpha)}$$

and

$$w(n) = \begin{cases} \text{some symmetric function with respect to} \\ \alpha \text{ over } 0 \leq n \leq M - 1 \\ 0, \text{ otherwise} \end{cases}$$
